



## A+ LAB SERIES V5

### Lab 13: Managing Storage

Document Version: **2026-01-02**

| Material in this Lab Aligns to the Following |                                                                                                          |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------|
| CompTIA A+ (220-1201) Exam Objectives        | 3.4: Compare and contrast storage devices.<br>5.2: Given a scenario, troubleshoot drive and RAID issues. |
| CompTIA A+ (220-1202) Exam Objectives        | 1.4: Given a scenario, use Microsoft Windows operating system features and tools.                        |

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## Introduction

The most common and most catastrophic hardware failures in computers are caused by hard disk drive issues. Even if proper backup procedures are followed, data may still be lost (including any changes since the last backup), and productivity may be reduced while the computer's hard drives are replaced.

One solution to this problem was the development of *RAID (Redundant Array of Independent or Inexpensive Disks)* protocols. These protocols allow the collection of physical drives into a logical "array," with the operating system treating the array as a single logical drive. Then the data is redundantly saved to more than one drive so that, if one (and in some cases 2 or 3) drive fails, the data remains intact and the computer will continue to run until the failed drive is replaced.

There are several "levels" of RAID that are in common use today:

|                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RAID0<br>(Striping)                                          | In this RAID level, multiple drives are added to the array and are treated as a single logical drive (the operating system sees it as a single file system). However, this is NOT a redundant array (despite the name) because if one drive in the array fails, the entire array is lost.                                                                                                                                    |
| RAID1<br>(Mirroring)                                         | At this RAID level, pairs of drives are assembled into an array and treated as a single logical drive. When data is written, it is written to both drives simultaneously. If one of the drives fails, the data is still intact on the other drive.                                                                                                                                                                           |
| RAID5<br>(Spanning with Parity)                              | In this RAID level, three or more drives are built into an array and are treated as a single logical drive. Unlike RAID0, a portion of each drive [1/# of drives ... 3 drives 1/3, 4 drives 1/4, 5 drives 1/5, and so on] is reserved for "parity." If one drive fails, the parity of the other drives will "make up" the data that was on the failed drive.                                                                 |
| Other RAID Levels (less commonly used)                       |                                                                                                                                                                                                                                                                                                                                                                                                                              |
| RAID6 & RAID7<br>(Spanning with double or triple parity)     | In these two RAID levels, four or more drives (RAID6) and five or more drives (RAID7) are treated as a single logical drive. Like RAID5, a portion of each drive is reserved for parity. The difference is that RAID6 supports double parity and can keep going if two drives in the array fail, and RAID7 supports up to a three-drive failure.                                                                             |
| RAID0+1                                                      | At this RAID level, we combine RAID0 (striping) and RAID1 (mirroring). It requires at least four drives in the array. Two or more drives are built into a RAID0 array, and the other two or more drives are built into a second RAID0 array. Finally, the two RAID0 arrays are combined into a single RAID1 array. In this array configuration, if one entire RAID0 array fails, the entire system will continue to operate. |
| RAID10<br>(Do not refer to it as RAID "ten" but as one-zero) | At this RAID level, we combine RAID0 (striping) and RAID1 (mirroring). It requires at least four drives in the array. Pairs of drives are built into a RAID1 array, and the RAID1 arrays are combined into a single RAID0 array. In this array configuration, one drive could fail in any or all of the RAID 1 arrays, and the entire system will continue to operate. This RAID level is more commonly used than RAID0+1.   |

Starting with Windows 8, Microsoft added a new disk configuration called *Storage Spaces*. *Storage Spaces* is a new *Microsoft* take on *RAID*. One of the major drawbacks of *RAID* (especially *RAID5*, *RAID6*, and *RAID7*) is performance, especially when the *RAID* function is implemented in software rather than by dedicated *RAID* hardware controllers. *Microsoft* has created an environment that provides similar redundancy, but does so in software. *Microsoft* claims that *Storage Spaces* pools are also more flexible than traditional *RAID* approaches.

- a. In this RAID level, three or more drives are built into an array and are treated as a single logical drive. Unlike RAID0, a portion of each drive [1/# of drives ... 3 drives 1/3, 4 drives 1/4, 5 drives 1/5, and so on] is reserved for “parity.” If one drive fails, the parity of the other drives will “make up” the data that was on the failed drive.
- b. In this level of RAID, we combine RAID0 (stripping) and RAID1 (mirroring). It requires a minimum of four drives in the array. Two or more drives are built into a RAID0 array, and the other two or more drives are built into a second RAID0 array. Finally, the two RAID0 arrays are combined into a single RAID1 array. In this form of an array, one entire RAID0 array could fail, and the entire system will continue to operate.
- c. In this RAID level, multiple drives are added to the array and are treated as a single logical drive (the operating system sees it as a single file system). However, this is NOT a redundant array (despite the name) because if one drive in the array fails, the entire array is lost.
- d. In this level of RAID, pairs of drives are built into an array and are treated as a single logical drive. When data is written, it is written to both drives simultaneously. If one of the drives fails, the data is still intact on the other drive.
- e. In this level of RAID, we combine RAID0 (stripping) and RAID1 (mirroring). It requires a minimum of four drives in the array. Pairs of drives are built into a RAID1 array, and the RAID1 arrays are combined into a single RAID0 array. In this form of an array, one drive could fail in any or all of the RAID1 arrays, and the entire system will continue to operate. This RAID level is more commonly used than RAID0+1.
- f. In these two levels of RAID, four or more (RAID6) and five or more (RAID7) drives are treated as a single logical drive. Like RAID5, a portion of each drive is reserved for parity. The difference is that RAID6 supports double parity and can keep going if two drives in the array fail, and RAID7 supports up to a three-drive failure.

## Objective

In this lab, we will be performing the following tasks:

- Use Computer Management and Disk Management to identify storage partitions.
- Use Disk Management to create a partition.
- Configure two drives to use RAID 1.
- Use Storage Spaces to help protect from drive failure.

## Lab Topology



## Lab Settings

The information in the table below is needed to complete the lab. The task sections below provide details on how to use this information.

| Virtual Machine | IP Address        | Account<br>(if needed) | Password<br>(if needed) |
|-----------------|-------------------|------------------------|-------------------------|
| Win10-OS        | 192.168.1.10 / 24 | sysadmin               | NDGLabpass123!          |
| Win11-OS        | 192.168.1.11 / 24 | sysadmin               | NDGLabpass123!          |

## 1 Managing Storage

### 1.1 Configuring RAID1 in Win11-OS

1. Open a console connection to the **Win11-OS** system.

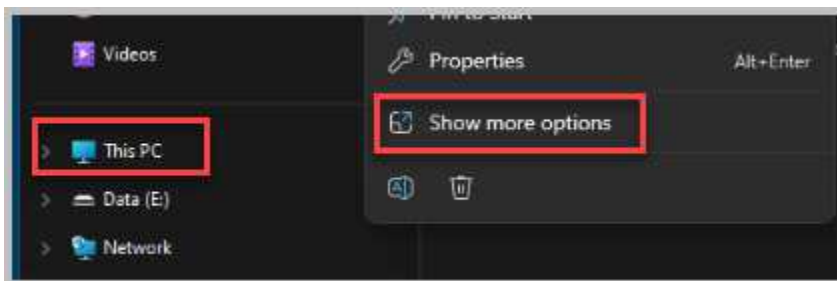


To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

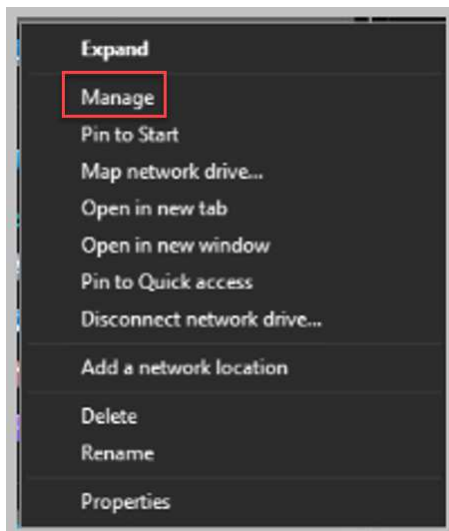
2. Select **File Explorer** on the taskbar.



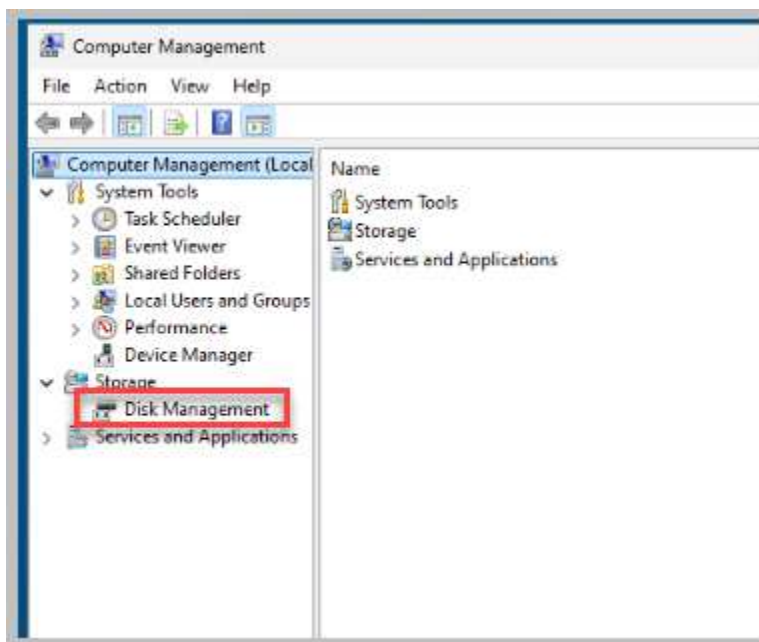
3. Right-click on the **This PC** icon and select **Show more options**.



4. Select **Manage**.

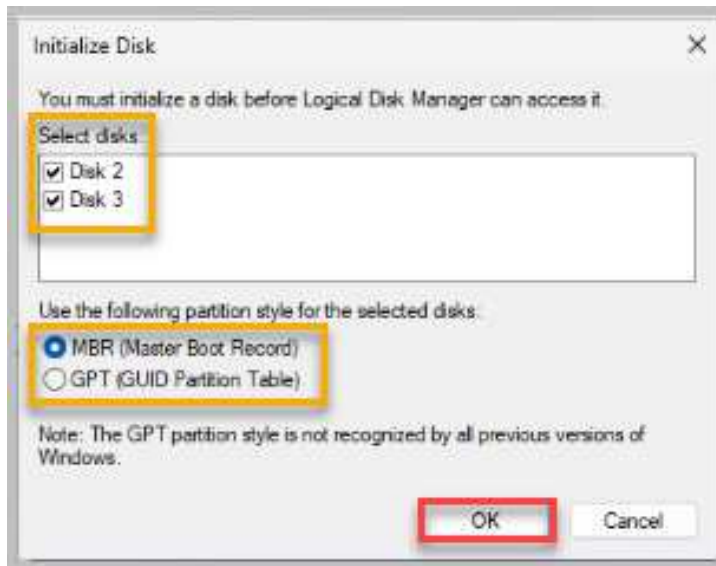


5. In the *Computer Management* window, click on **Disk Management** located in the left pane.



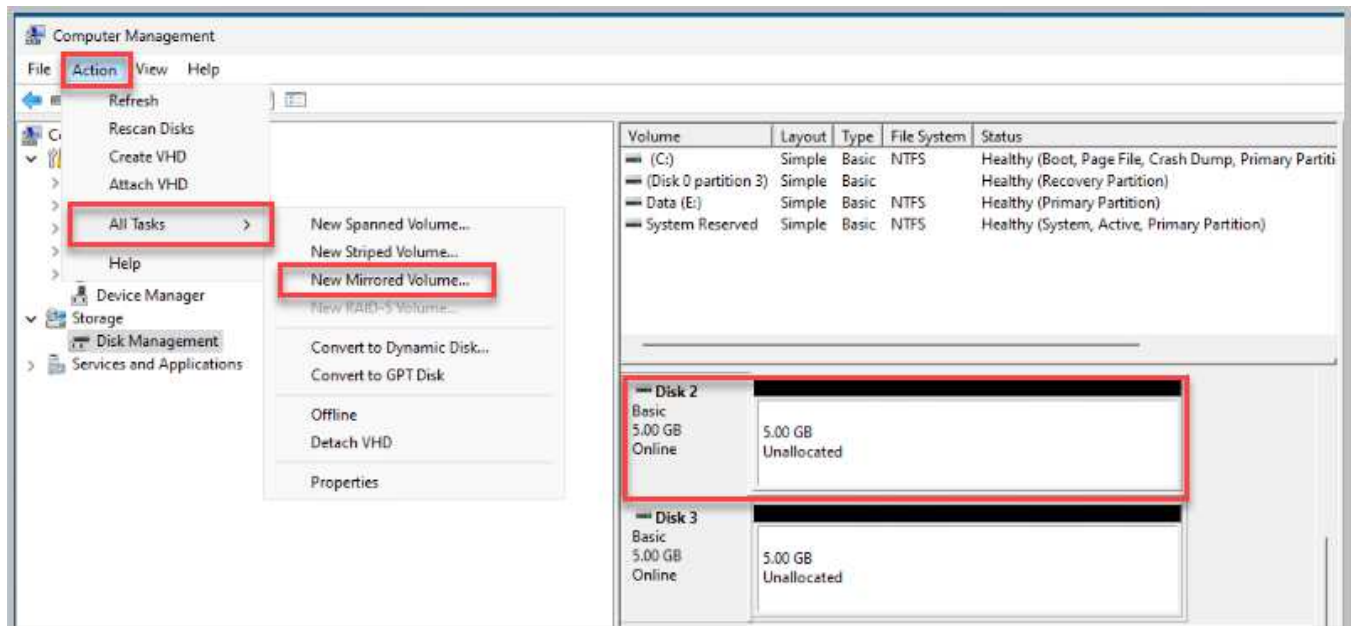


6. In the *Initialize Disk* pop-up window, verify that both *Disk 2 and Disk 3* are checked. Leave the default *MBR (Master Boot Record)* enabled and select **OK**.

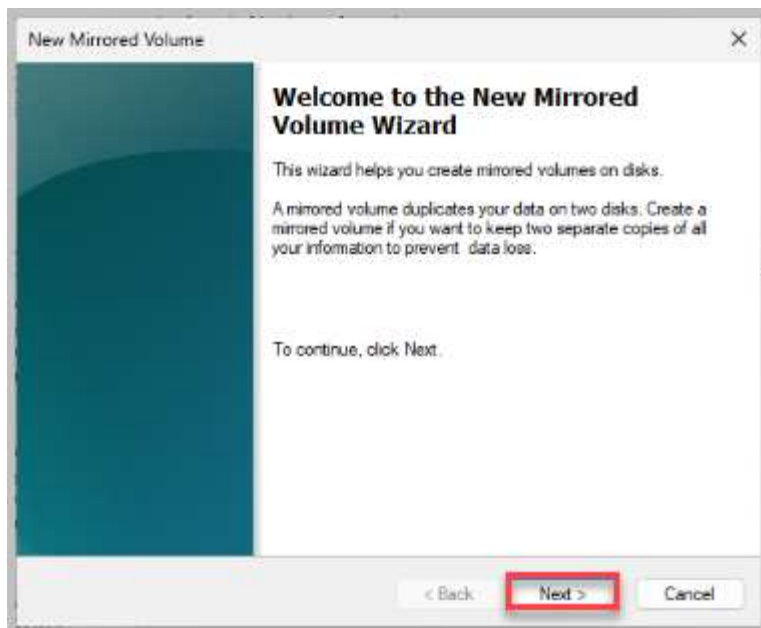


*MBR* (Master Boot Record) and *GPT* (GUID Partition Table) are two partitioning schemes with distinct features. *MBR* is compatible with older BIOS systems, supports up to 4 primary partitions, and can handle drives up to 2 TB. However, it stores data in a single location, making it more vulnerable to corruption. It's necessary for booting systems that use BIOS. In contrast, *GPT* is designed for newer UEFI systems, allowing up to 128 partitions and supporting drives up to 18 exabytes. *GPT* stores multiple copies of partition data with CRC checks for added reliability and integrity, making it more robust. It's essential for booting on UEFI systems. Generally, *GPT* is better suited for modern systems and larger drives, while *MBR* is ideal for older setups or smaller drives under 2 TB.

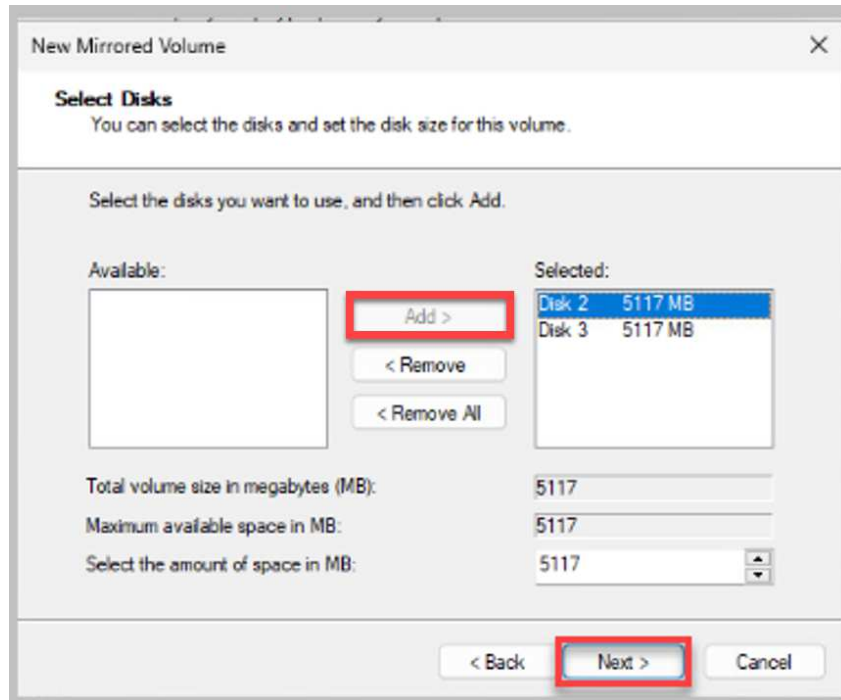
- Click on **Disk 2** from the list of drives available to highlight it and then click on the **Action** menu item, selecting **All Tasks > New Mirrored Volume**.



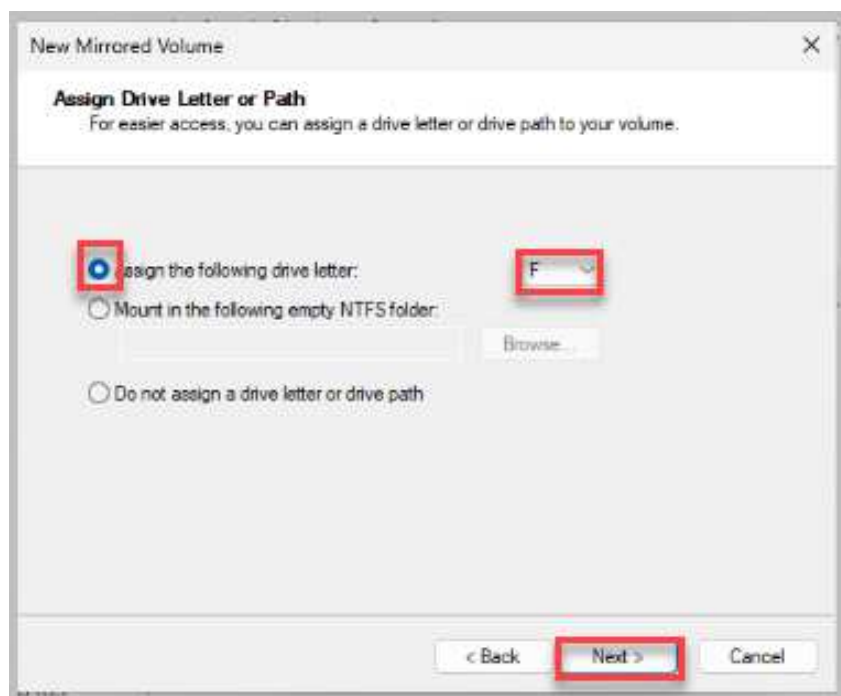
- Notice the *New Mirrored Volume* wizard appears. Click **Next**.



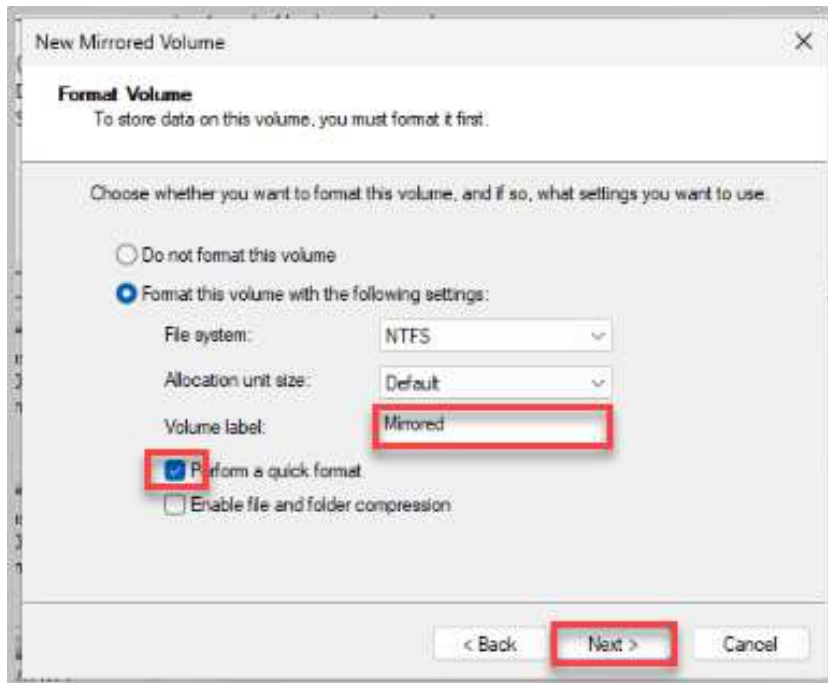
9. The first step is to select the drives that will be included in a *RAID1* format. *Disk 2* should already be in the *Selected* pane. Click *Disk 3*, then click *Add >* to move it from *Available* to *Selected*. Verify that both *Disk 2* and *Disk 3* are in the *Selected* field. Click **Next**.



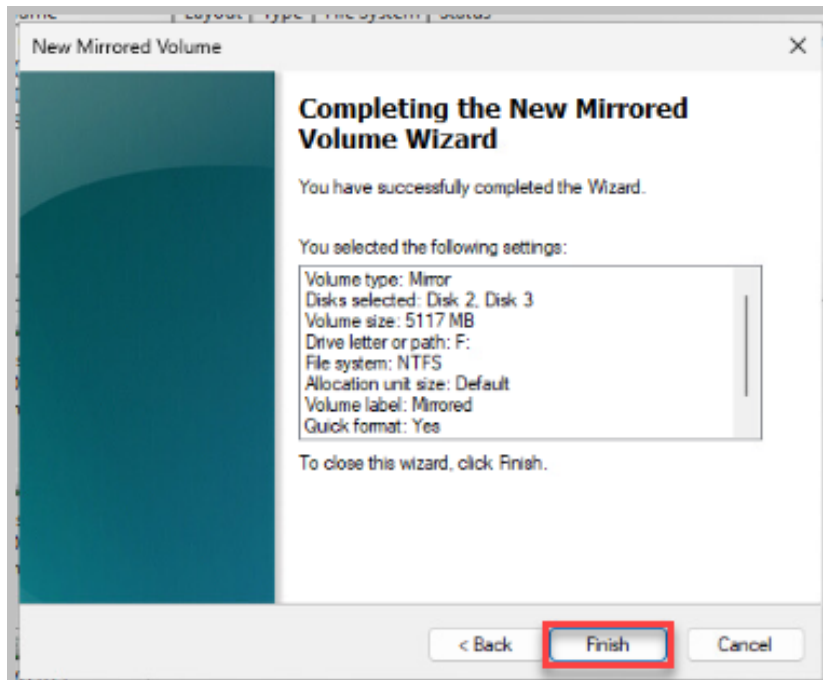
10. At this point, the mirrored array is just another file system for *Windows* that will be treated just like any other partition. Ensure that the radio button for **Assign the following drive letter** is selected and pick **F** from the dropdown menu to be chosen as the drive letter. Click **Next**.



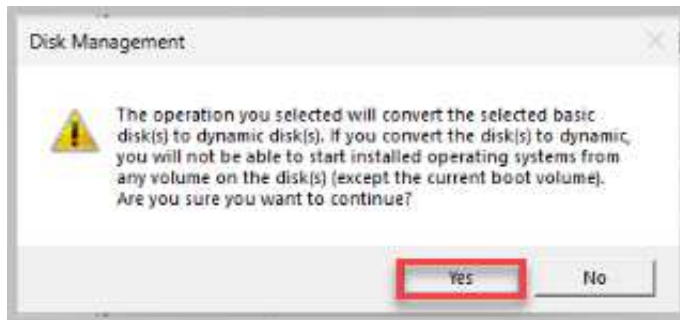
11. On the *Format Volume* step, type **mirrored** in the *Volume label* text field and **check** the checkbox for **Perform a quick format**. Click **Next**.



12. Confirm the changes and click the **Finish** button.



13. If prompted with a disk management warning, click **Yes** to convert a “basic” disk to a “dynamic” disk.



14. Notice *Disk 2* and *3* changed in the list. They are now both in the *Mirrored* array and reporting back as *healthy*. Close the **Computer Management** window.



15. If it is not open, select **File Explorer** on the taskbar.



- Click on the **This PC** icon. In the *Devices and drives* pane, notice the new *Mirrored* volume is listed ready to be used.



- Close the **File Explorer** window.
- Leave the *Win11-OS* window open to continue with the next task.

## 1.2 Deleting Mirror Drives and Volumes in Win11-OS

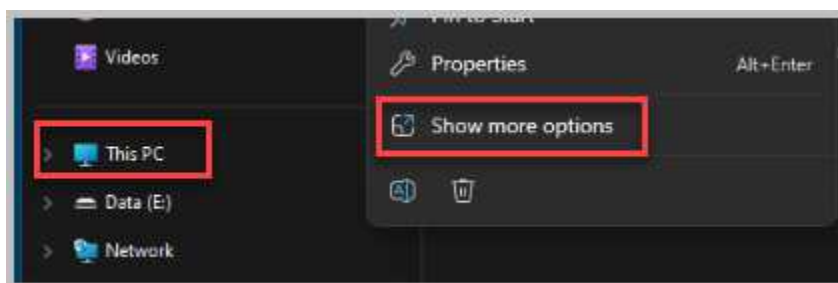


**Delete the mirrored drives 2 and 3, including the associated volume.** This step is crucial for the successful completion of Activity 1.3.

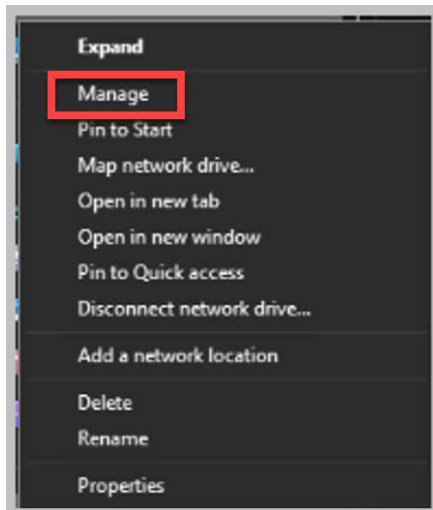
- Select **File Explorer** on the taskbar.



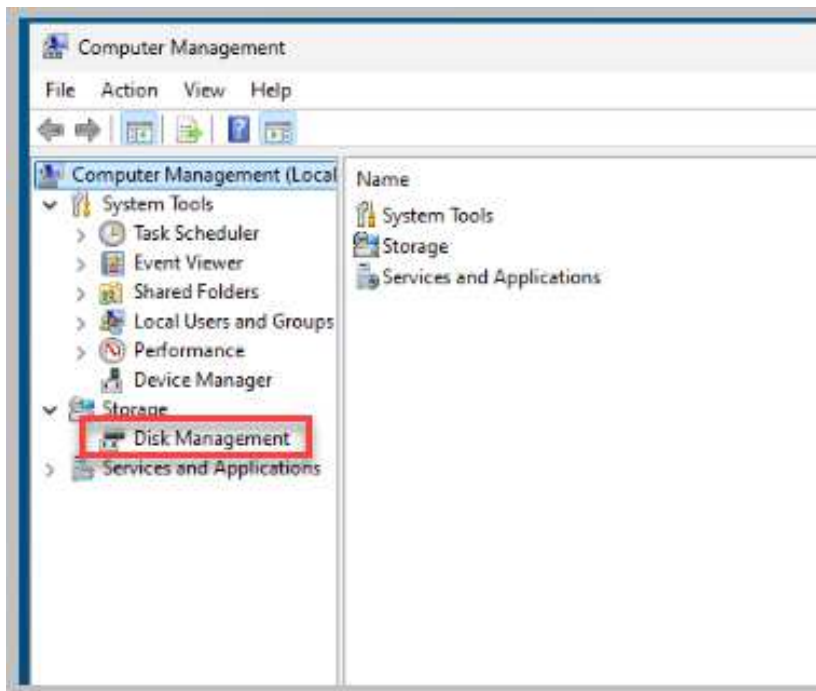
- Right-click on the **This PC** icon and select **Show more options**.



3. Select **Manage**.



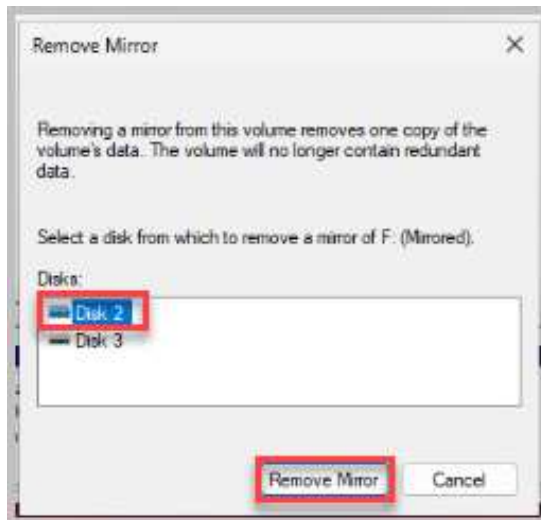
4. In the *Computer Management* window, click on **Disk Management** located in the left pane.



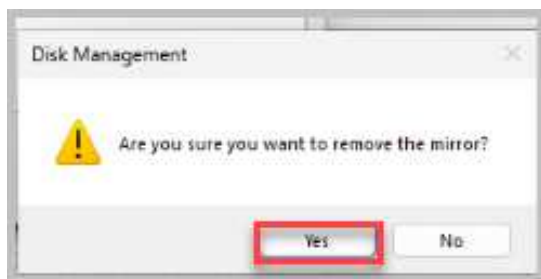
5. Right-click **Disk 2** and select **Remove Mirror...**



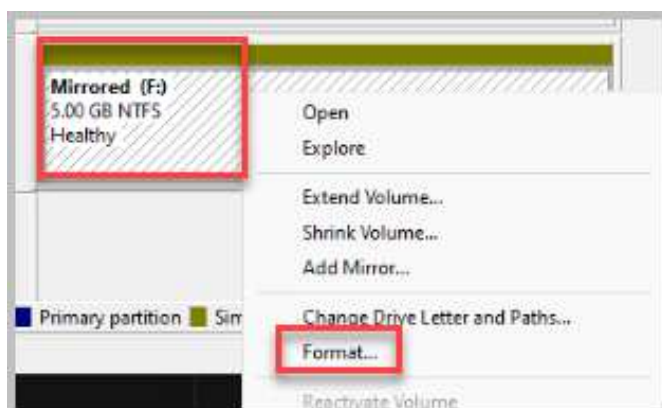
6. In the *Remove Mirror* notification window, select **Disk 2** and select **Remove Mirror**.



7. In the *Disk Management* notification window, select **Yes** to remove the mirror.

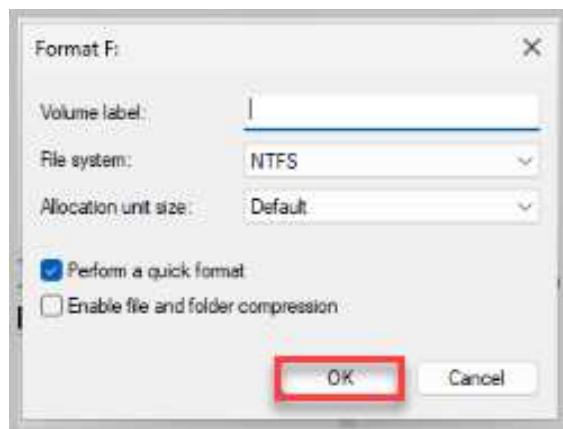


8. As *Disk 2* was removed from the *Mirror Volume*, *Disk 3* is still listed as a member of the *Mirror Volume*. Right click **Mirrored (F:)** and select **Format**.





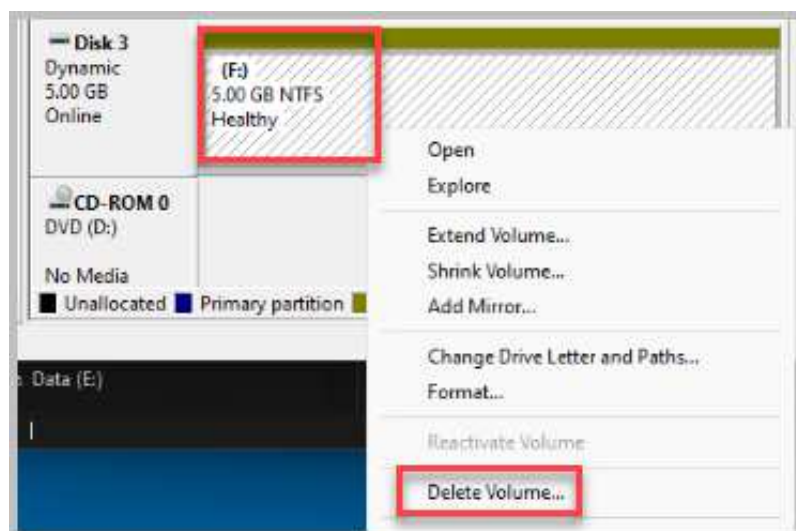
9. Delete the *Volume label* and select **OK**.



10. Select **OK** to format *F Volume*.



11. Right click the *Volume F* and select **Delete Volume**.



12. Select **Yes** to *delete simple volume*.



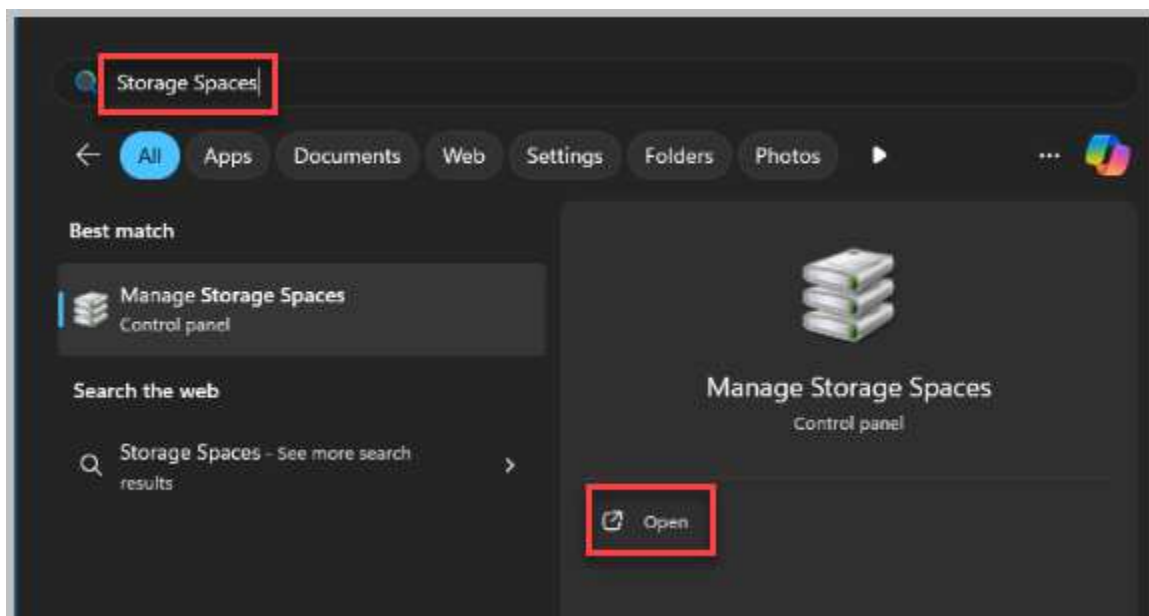
13. Close all open windows. Leave the *Win11-OS* console open.

### 1.3 Configuring Storage Spaces in Win11-OS

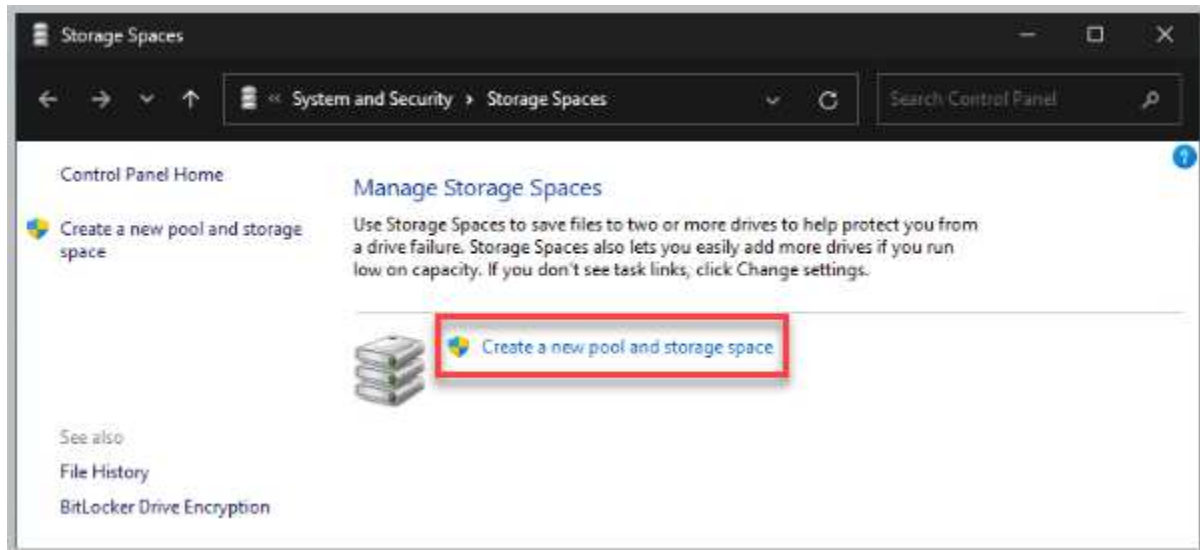
1. On the *Win11-OS* desktop, click on the **Start** button.



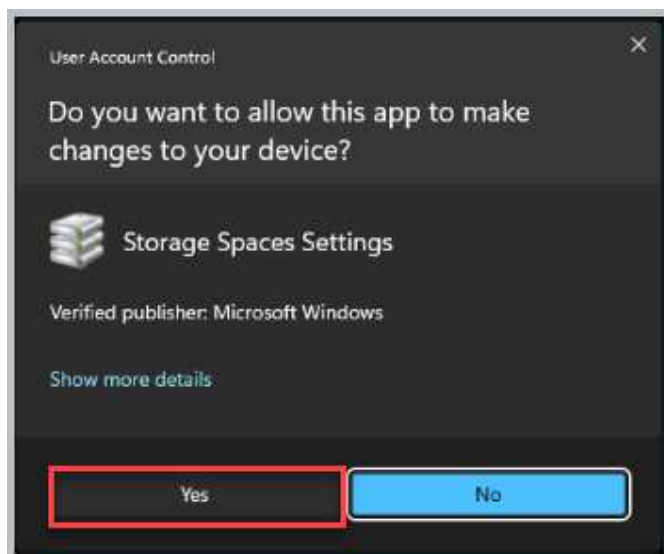
2. In the search field, type **Storage Spaces**. From the search results, select **Open** for **Manage Storage Spaces**.



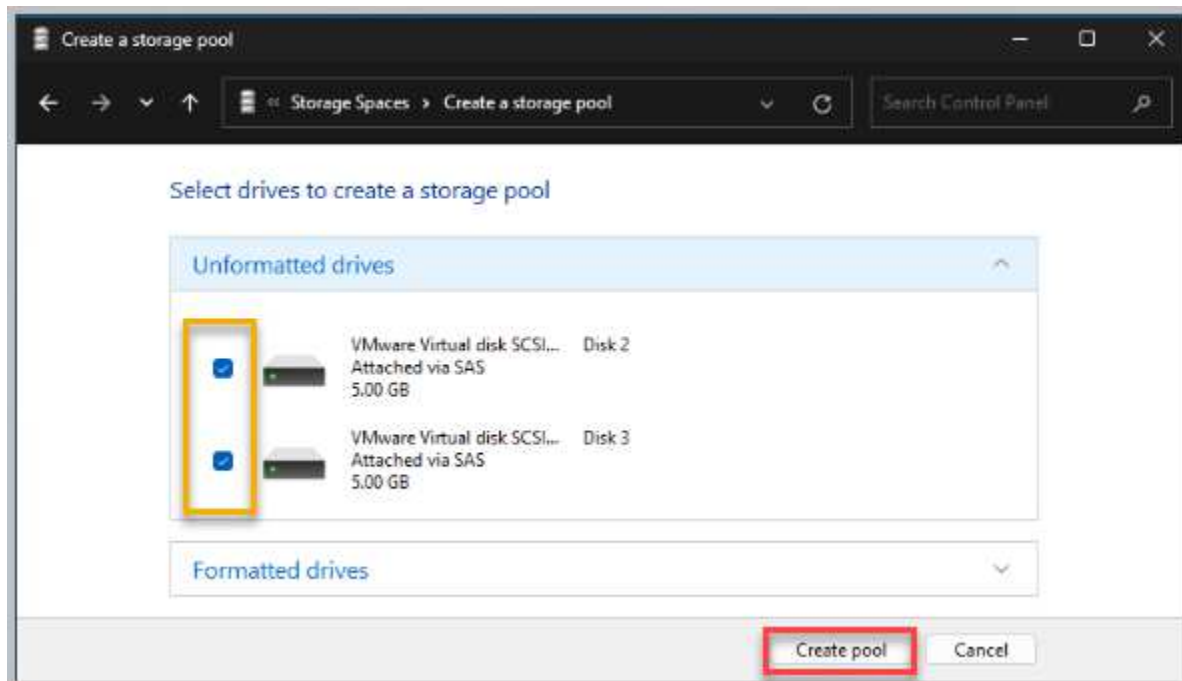
3. In the *Storage Spaces* window, click on **Create a new pool and storage space**.



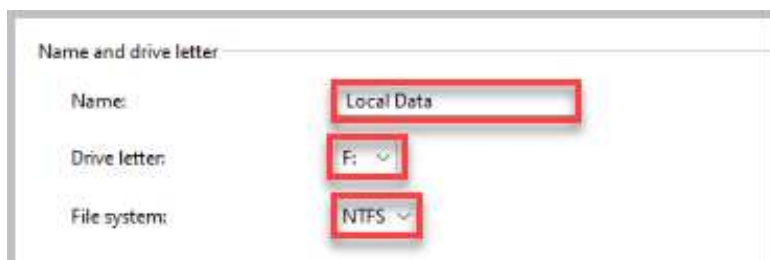
4. When prompted with User Account Control, click **Yes** to continue.



5. The next step is to select which drives to be included in the *Storage Pool*. Make sure **Disk 2** and **3** are selected and click on the **Create pool** button.



6. The last step is to create the *Storage Space* from the pool. Type **Local Data** in the *Name* text field. Choose **F:** from the *Drive letter* dropdown menu. Leave **NTFS** selected.



7. The *Resiliency* section allows you to select the method to organize the space and how to protect your data (think of it as the *RAID* section). Choose **Two-Way Mirror** from the *Resiliency type* dropdown menu.



Below is a table explaining each of the options:

|                               |                                                                                                                                                                                                                                                                                                                          |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Simple (no resiliency)</b> | This is the <i>Storage Space</i> version of <i>RAID0</i> . You can start with one drive and add more as needed. As with <i>RAID0</i> , if one drive in the pool fails, all data is lost. The pool size is the sum of all the drives.                                                                                     |
| <b>Two-Way Mirror</b>         | This is the <i>Storage Space</i> version of <i>RAID1</i> . You must start with at least two drives. During a write action, data is written to both drives. In the event of a single drive failure, the data will still be available on the second drive. The pool's total size is half the total size of all the drives. |
| <b>Three-Way Mirror</b>       | This is the <i>Storage Space</i> version of <i>RAID6</i> and <i>RAID10</i> . Requires at least five drives and can protect from the failure of two drives in the pool. The pool's total size is one-third of the total size of all the drives; the other two-thirds is used for parity.                                  |
| <b>Parity</b>                 | This is the <i>Storage Space</i> version of <i>RAID5</i> . It requires at least three drives and can protect from the failure of one drive. The pool's total size is two-thirds of the total size of all the drives, and the remaining one-third is used for parity.                                                     |

8. The *Size* section allows you to change the size. The default value depends on the *chosen Resiliency Type*. You can increase the size in this type of setup, even if you don't have the capacity. Later, when the actual space in the pool runs out, you can add additional drives to the pool without provisioning the *Storage Space* again. Enter **4 GB** in the *Size (maximum)* field.



Size

Total pool capacity: 8.74 GB

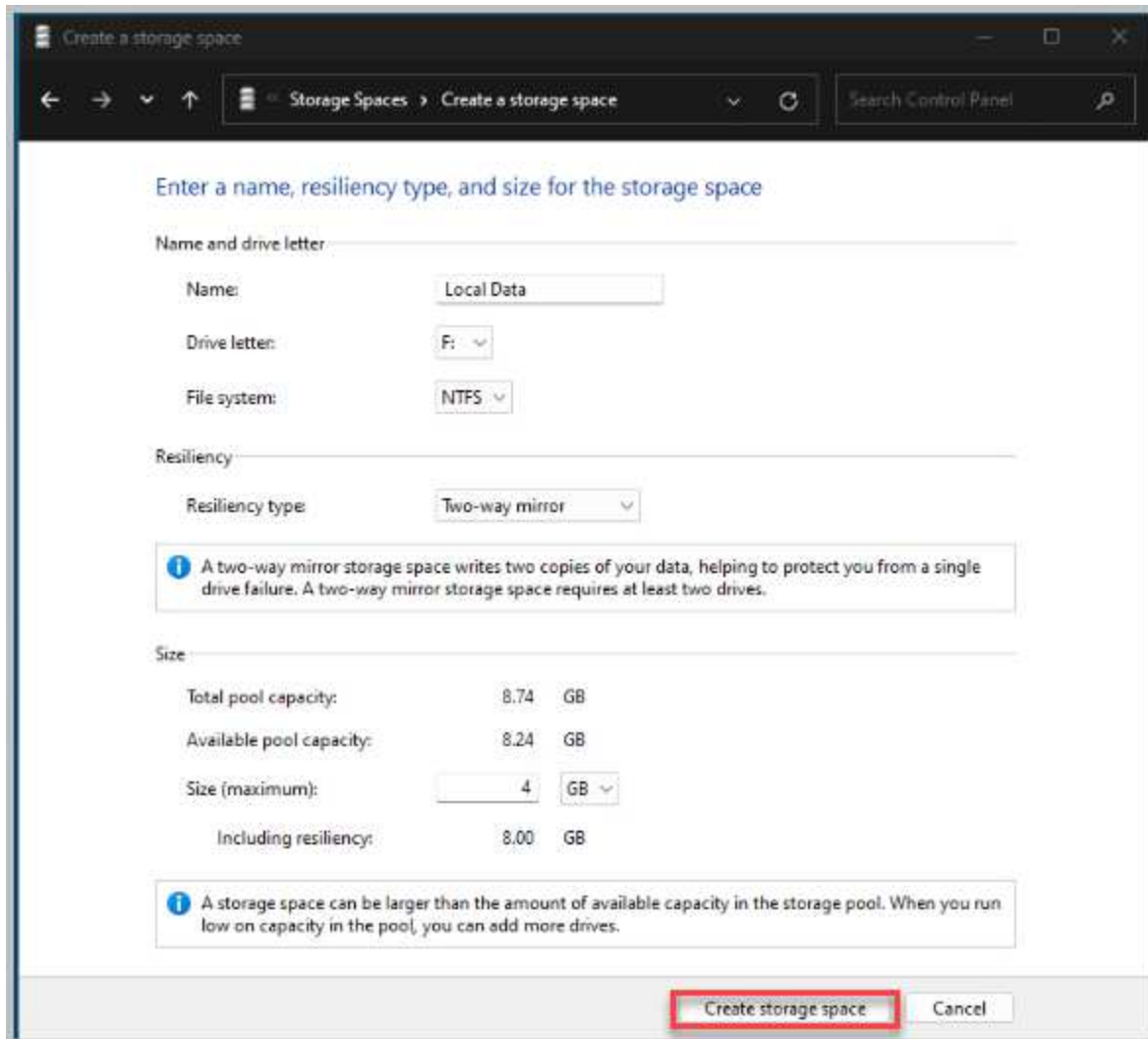
Available pool capacity: 8.24 GB

Size (maximum): 4 GB

Including resiliency: 8.00 GB

*A storage space can be larger than the amount of available capacity in the storage pool. When you run low on capacity in the pool, you can add more drives.*

9. Review your proposed configuration and click **Create storage space**.



Create a storage space

Storage Spaces > Create a storage space

Enter a name, resiliency type, and size for the storage space

Name and drive letter

Name: Local Data

Drive letter: F: ▾

File system: NTFS ▾

Resiliency

Resiliency type: Two-way mirror ▾

**i** A two-way mirror storage space writes two copies of your data, helping to protect you from a single drive failure. A two-way mirror storage space requires at least two drives.

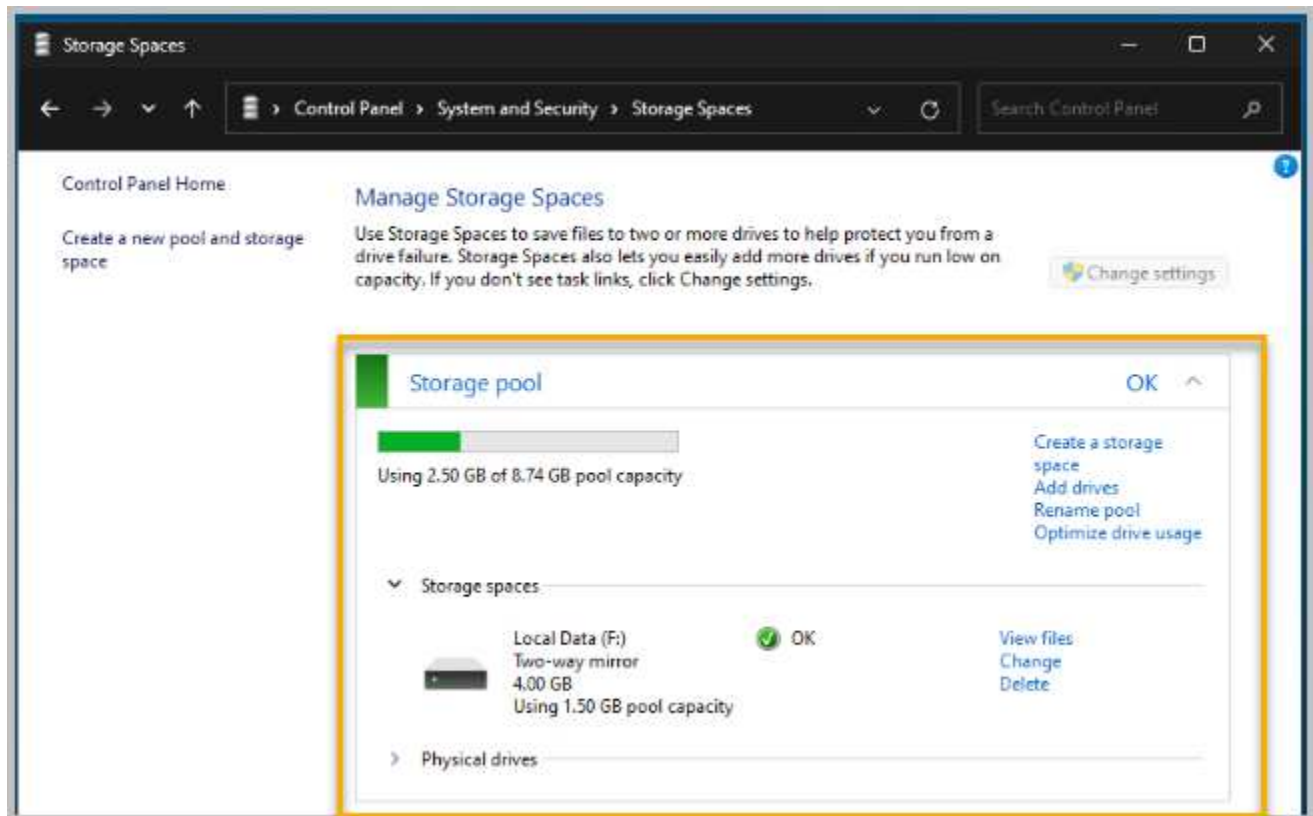
Size

|                          |      |      |
|--------------------------|------|------|
| Total pool capacity:     | 8.74 | GB   |
| Available pool capacity: | 8.24 | GB   |
| Size (maximum):          | 4    | GB ▾ |
| Including resiliency:    | 8.00 | GB   |

**i** A storage space can be larger than the amount of available capacity in the storage pool. When you run low on capacity in the pool, you can add more drives.

Create storage space Cancel

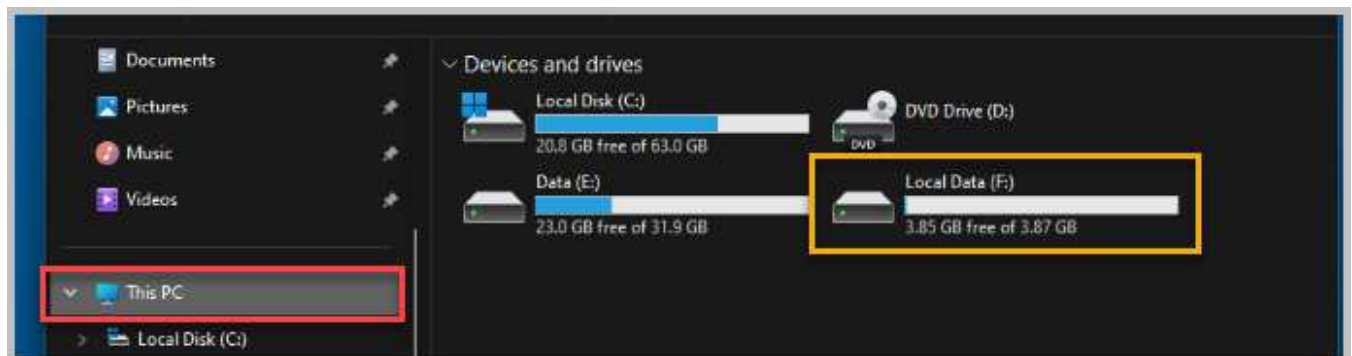
10. After the *Storage Pool* is formatted, the *Storage Spaces* management window will be displayed. Notice the new space is ready to be used. Close the **Storage Spaces** window.



11. Select **File Explorer** on the taskbar.



12. Click on the **This PC** icon and notice the new volume, *Local Data (F:)*, is listed.



13. The lab is now complete; you may end the reservation.